

VENNAM JAYA CHANDRA

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PROFESSIONAL SUMMARY

Integrated M.Tech Computer Science student (VIT-AP University) with hands-on experience building production-grade Machine Learning, Computer Vision, and NLP systems. Specializes in robust inference pipelines — from hybrid decision-fusion models (94.0% accuracy, 91.9% F1) to real-time 30 FPS gesture tracking and CNN-based surface defect inspection. Experienced in shipping containerized REST APIs with low latency and developing autonomous agentic AI workflows.

EDUCATION

Integrated M.Tech in Computer Science & Engineering — VIT-AP University

CGPA: 7.68 / 10.0 | Expected Graduation: June 2027 | Specialization: Machine Learning, Computer Vision, Agentic AI

TECHNICAL SKILLS

Languages: Python (Advanced), Java, SQL (MySQL), JavaScript, HTML5/CSS3

AI & Deep Learning: TensorFlow/Keras, PyTorch, OpenCV, MediaPipe, Scikit-learn, CNNs, Transfer Learning, Feature Engineering

NLP & Decision Systems: Tokenization, Subword Encoding, TF-IDF, Linear SVC, Logistic Regression, Explainable AI (XAI)

GenAI & Agentic AI: Hugging Face Agents (smolagents), Autonomous Agent Workflows, Tool Calling, RAG, Prompt Engineering

Backend & Cloud/Tools: Flask, RESTful APIs, MySQL Schema Design, Docker, Linux/Bash, Git/GitHub, Vercel, Render

FEATURED ENGINEERING PROJECTS

BullyMail Threat Intelligence — Hybrid Decision-Fusion NLP Engine [\[Live Demo ↗\]](#) [\[GitHub ↗\]](#) | *Python, Scikit-learn, Linear SVC, TF-IDF, REST APIs, Live IMAP*

- Architected a Hybrid Decision-Fusion NLP engine combining statistical ML (TF-IDF + Linear SVC) with a weighted 5-tier threat taxonomy and anti-adversarial text normalization to detect abusive content in live email streams.
- Benchmarked on a 10,000-sample academic email corpus (75/25 stratified split), achieving 94.00% accuracy, 100.0% precision, and a 91.89% F1-score, boosting threat recall to 85.0% (+47.2% gain over baseline).
- Engineered sub-second inference latency prioritized over heavy Transformers (BERT/RoBERTa) to sustain continuous live IMAP mailbox ingestion (validated on 271 real messages) with token-level attribution (XAI).
- Deployed production application on Render featuring real-time mailbox synchronization, automated risk scoring, and interactive threat-classification dashboards.

SkyWrite — Real-Time AI Gesture-Controlled Virtual Whiteboard [\[GitHub ↗\]](#) | *Python, OpenCV, MediaPipe, TensorFlow/Keras, CNN*

- Engineered a touchless air-writing system tracking 21 3D hand landmarks at 30 FPS on standard laptop webcams with zero perceptible latency using an optimized OpenCV and MediaPipe processing pipeline.
- Developed a 4-state finite state machine recognizing distinct gestures: Index-finger drawing, Two-finger navigation & palette picker, Thumb-index pinch eraser, and Pinky-swipe canvas clear.
- Trained a custom TensorFlow/Keras CNN on the EMNIST dataset to classify and digitize continuous air-drawn stroke coordinates into real-time alphanumeric characters.
- Eliminated stroke breaks and tracking dropouts by designing an exponential moving average (EMA) coordinate smoothing filter and spatial boundary thresholding, reducing jitter by over 80%.

Full-Stack Fabric Marketplace with AI Defect Detection [\[Live Demo ↗\]](#) [\[GitHub ↗\]](#) | *Flask, TensorFlow/Keras, OpenCV, MySQL, REST APIs, Docker*

- Built a full-stack e-commerce inspection platform featuring 3-tier Role-Based Access Control (Admin, Seller, Buyer), normalized MySQL schemas, and RESTful APIs for automated inventory auditing.
- Engineered a dual-stage vision defect pipeline: an OpenCV contour-detection engine to isolate anomalous surface geometries (cuts/holes) paired with a custom TensorFlow/Keras CNN for binary classification (Good vs. Defect Cloth).
- Implemented automated batch image inspection alongside a frame-by-frame video inference pipeline to flag anomalous surface patterns on moving textile rolls in production.
- Integrated automated seller catalog-gating workflows, preventing defective product indexing prior to listing submission and automating vendor quality compliance.

CERTIFICATIONS & TECHNICAL CREDENTIALS

- Hugging Face Agents Course Certification:** Practical autonomous agent architectures, tool-calling APIs, and multi-agent system design.
- Oracle Cloud Infrastructure (OCI) 2025 AI Foundations Associate:** Enterprise cloud architectures, ML deployment pipelines, and AI security.
- Microsoft & LinkedIn Generative AI / Google Gemini Academy:** Advanced prompt engineering, multimodal architectures, and LLM application design.